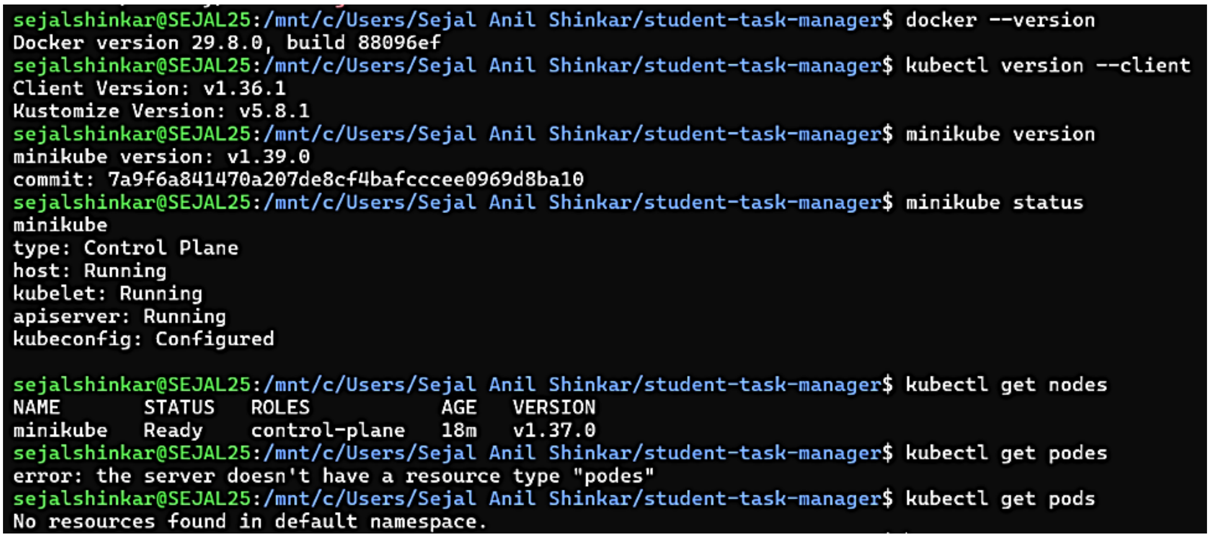
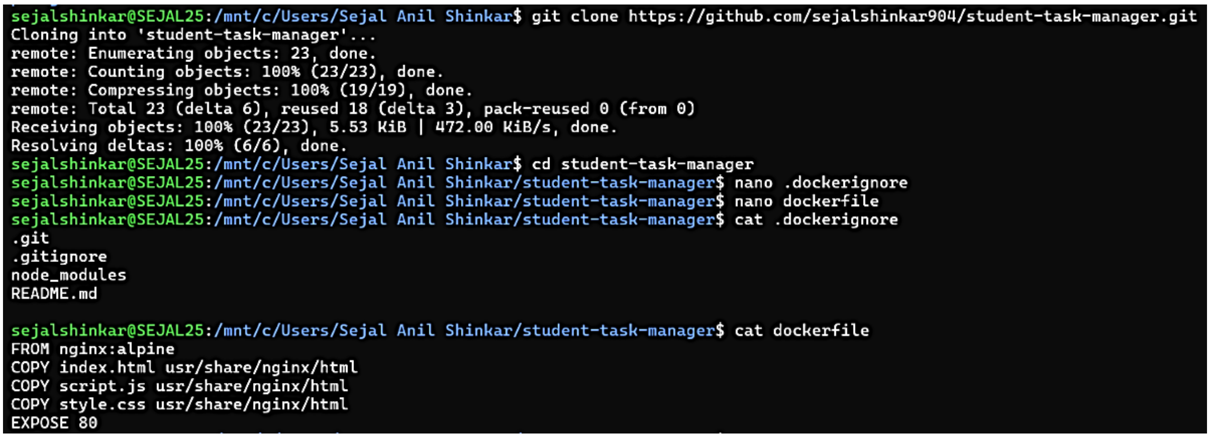
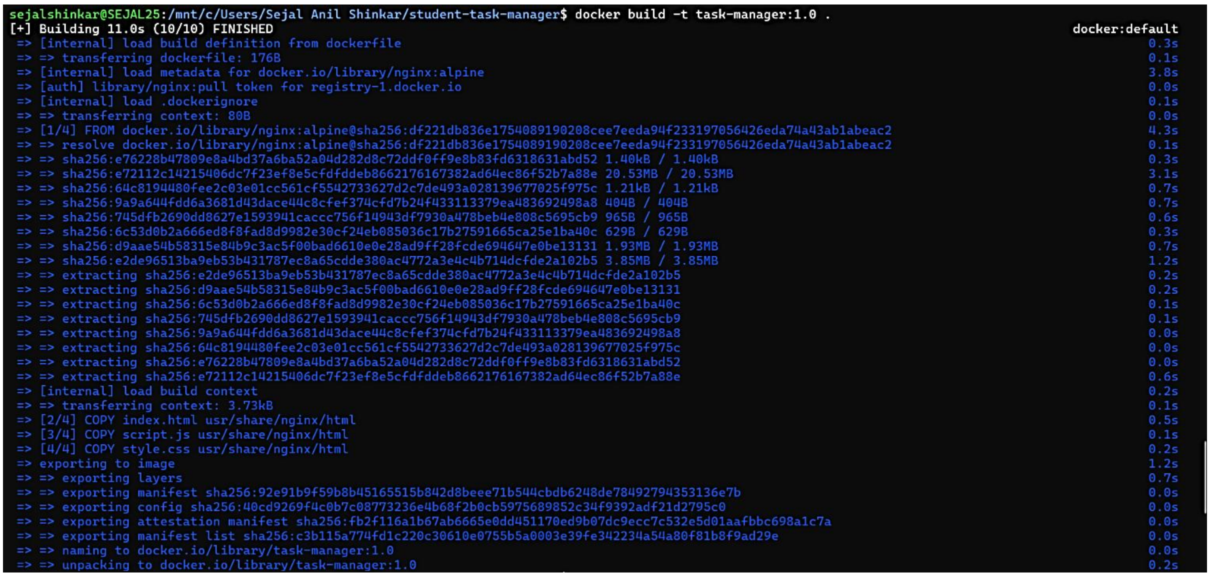


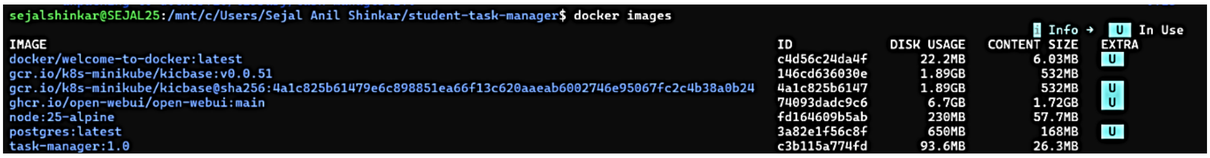
```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ docker --version
Docker version 29.8.0, build 88096ef
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl version --client
Client Version: v1.36.1
Kustomize Version: v5.8.1
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ minikube version
minikube version: v1.39.0
commit: 7a9f6a841470a207de8cf4bafcccee0969d8ba10
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   18m   v1.37.0
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl get podes
error: the server doesn't have a resource type "podes"
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl get pods
No resources found in default namespace.
```



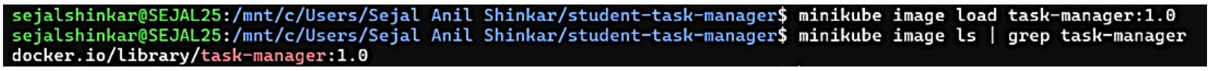
```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar$ git clone https://github.com/sejalshinkar904/student-task-manager.git
Cloning into 'student-task-manager'...
remote: Enumerating objects: 23, done.
remote: Counting objects: 100% (23/23), done.
remote: Compressing objects: 100% (19/19), done.
remote: Total 23 (delta 6), reused 18 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (23/23), 5.53 KiB | 472.00 KiB/s, done.
Resolving deltas: 100% (6/6), done.
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar$ cd student-task-manager
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ nano .dockerignore
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ nano dockerfile
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ cat .dockerignore
.git
.gitignore
node_modules
README.md

sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ cat dockerfile
FROM nginx:alpine
COPY index.html usr/share/nginx/html
COPY script.js usr/share/nginx/html
COPY style.css usr/share/nginx/html
EXPOSE 80
```



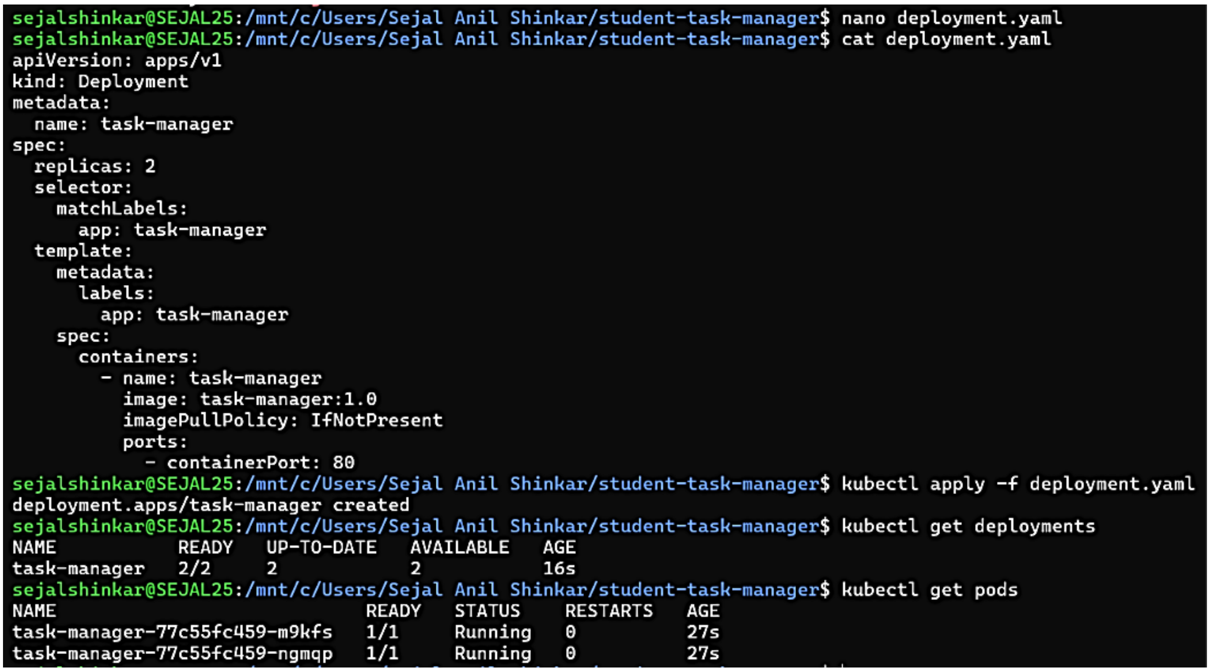
```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ docker build -t task-manager:1.0 .
[+] Building 11.0s (10/10) FINISHED                                                                   docker:default
 => [internal] load build definition from dockerfile                                                            0.3s
 => => transferring dockerfile: 176B                                                                            0.1s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                 3.8s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                    0.0s
 => [internal] load .dockerignore                                                                               0.1s
 => => transferring context: 80B                                                                                0.0s
 => [1/4] FROM docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2  4.3s
 => => resolve docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2  0.1s
 => => sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72ddf0ff9e8b83fd6318631abd52 1.40kB / 1.40kB                  0.3s
 => => sha256:e72112c14215406dc7f23ef8e5cfdfddeb8662176167382ad64ec86f52b7a88e 20.53MB / 20.53MB                3.1s
 => => sha256:64c8194480fee2c03e01cc561cf5542733627d2c7de493a028139677025f975c 1.21kB / 1.21kB                  0.7s
 => => sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f433113379ea483692498a8 404B / 404B                      0.7s
 => => sha256:745dfb2690dd8627e1593941caccc756f14943df7930a478beb4e808c5695cb9 965B / 965B                      0.6s
 => => sha256:6c53d0b2a666ed8f8fad8d9982e30cf24eb085036c17b27591665ca25e1ba40c 629B / 629B                      0.3s
 => => sha256:d9aae54b58315e84b9c3ac5f00bad6610e0e28ad9ff28fcde694647e0be13131 1.93MB / 1.93MB                  0.7s
 => => sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3e4c4b714dcfde2a102b5 3.85MB / 3.85MB                  1.2s
 => => extracting sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3e4c4b714dcfde2a102b5                       0.2s
 => => extracting sha256:d9aae54b58315e84b9c3ac5f00bad6610e0e28ad9ff28fcde694647e0be13131                       0.2s
 => => extracting sha256:6c53d0b2a666ed8f8fad8d9982e30cf24eb085036c17b27591665ca25e1ba40c                       0.1s
 => => extracting sha256:745dfb2690dd8627e1593941caccc756f14943df7930a478beb4e808c5695cb9                       0.1s
 => => extracting sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f433113379ea483692498a8                       0.0s
 => => extracting sha256:64c8194480fee2c03e01cc561cf5542733627d2c7de493a028139677025f975c                       0.0s
 => => extracting sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72ddf0ff9e8b83fd6318631abd52                       0.0s
 => => extracting sha256:e72112c14215406dc7f23ef8e5cfdfddeb8662176167382ad64ec86f52b7a88e                       0.6s
 => [internal] load build context                                                                               0.2s
 => => transferring context: 3.73kB                                                                             0.1s
 => [2/4] COPY index.html usr/share/nginx/html                                                                  0.5s
 => [3/4] COPY script.js usr/share/nginx/html                                                                   0.1s
 => [4/4] COPY style.css usr/share/nginx/html                                                                   0.2s
 => exporting to image                                                                                          1.2s
 => => exporting layers                                                                                         0.7s
 => => exporting manifest sha256:92e91b9f59b8b45165515b842d8beee71b544cbdb6248de78492794353136e7b                0.0s
 => => exporting config sha256:40cd9269f4c0b7c08773236e4b68f2b0cb5975689852c34f9392adf21d2795c0                  0.0s
 => => exporting attestation manifest sha256:fb2f116a1b67ab6665e0dd451170ed9b07dc9ecc7c532e5d01aafbbc698a1c7a    0.0s
 => => exporting manifest list sha256:c3b115a774fd1c220c30610e0755b5a0003e39fe342234a54a80f81b8f9ad29e           0.0s
 => => naming to docker.io/library/task-manager:1.0                                                             0.0s
 => => unpacking to docker.io/library/task-manager:1.0                                                          0.2s
```



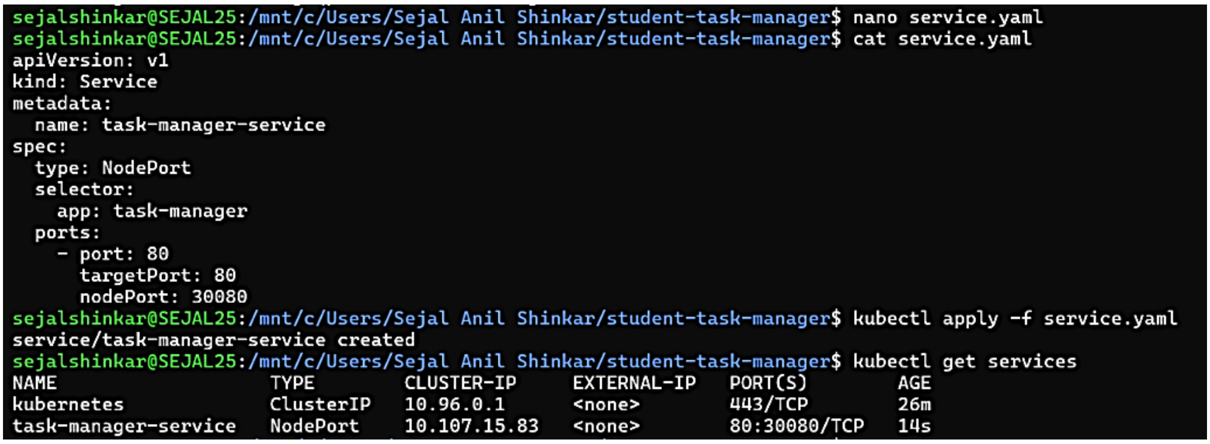
```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ docker images
                                                                                                        i Info →   U  In Use
IMAGE                                                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest                                                      c4d56c24da4f       22.2MB         6.03MB    U
gcr.io/k8s-minikube/kicbase:v0.0.51                                                  146cd636030e       1.89GB          532MB
gcr.io/k8s-minikube/kicbase@sha256:4a1c825b61479e6c898851ea66f13c620aaeab6002746e95067fc2c4b38a0b24  4a1c825b6147  1.89GB   532MB    U
ghcr.io/open-webui/open-webui:main                                                   74093dadc9c6        6.7GB         1.72GB    U
node:25-alpine                                                                       fd164609b5ab        230MB         57.7MB
postgres:latest                                                                      3a82e1f56c8f        650MB          168MB    U
task-manager:1.0                                                                     c3b115a774fd       93.6MB         26.3MB
```



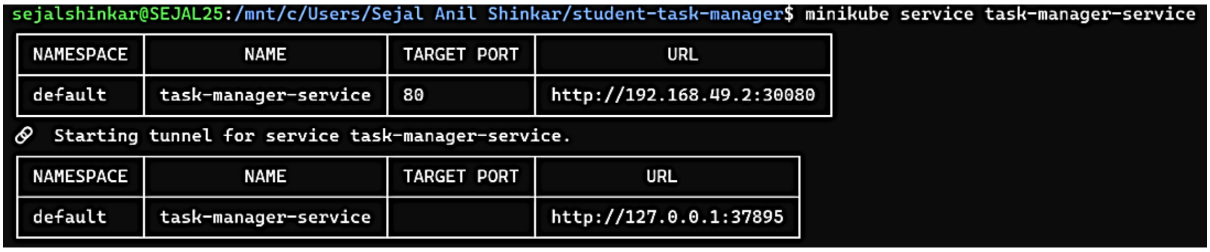
```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ minikube image load task-manager:1.0
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ minikube image ls | grep task-manager
docker.io/library/task-manager:1.0
```



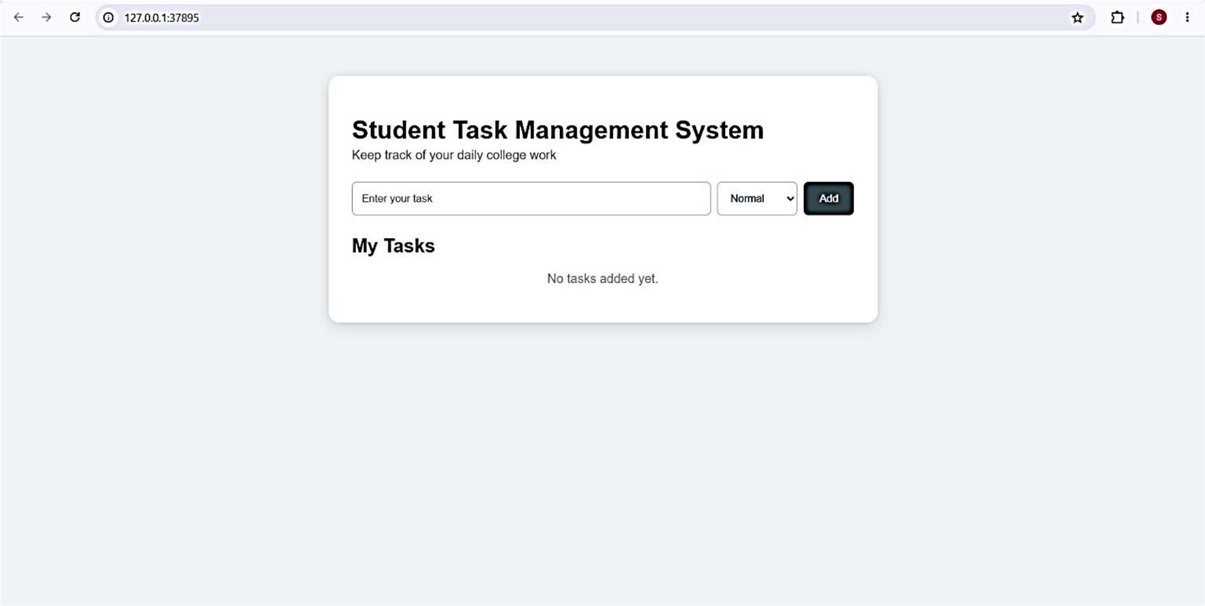
```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ nano deployment.yaml
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: task-manager
spec:
  replicas: 2
  selector:
    matchLabels:
      app: task-manager
  template:
    metadata:
      labels:
        app: task-manager
    spec:
      containers:
        - name: task-manager
          image: task-manager:1.0
          imagePullPolicy: IfNotPresent
          ports:
            - containerPort: 80
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl apply -f deployment.yaml
deployment.apps/task-manager created
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl get deployments
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
task-manager   2/2     2            2           16s
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl get pods
NAME                            READY   STATUS    RESTARTS   AGE
task-manager-77c55fc459-m9kfs   1/1     Running   0          27s
task-manager-77c55fc459-ngmqp   1/1     Running   0          27s
```



```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ nano service.yaml
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: task-manager-service
spec:
  type: NodePort
  selector:
    app: task-manager
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl apply -f service.yaml
service/task-manager-service created
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ kubectl get services
NAME                   TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes             ClusterIP   10.96.0.1      <none>        443/TCP        26m
task-manager-service   NodePort    10.107.15.83   <none>        80:30080/TCP   14s
```



```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar/student-task-manager$ minikube service task-manager-service
|-----------|----------------------|-------------|---------------------------|
| NAMESPACE |         NAME         | TARGET PORT |            URL            |
|-----------|----------------------|-------------|---------------------------|
| default   | task-manager-service | 80          | http://192.168.49.2:30080 |
|-----------|----------------------|-------------|---------------------------|
🔗  Starting tunnel for service task-manager-service.
|-----------|----------------------|-------------|------------------------|
| NAMESPACE |         NAME         | TARGET PORT |          URL           |
|-----------|----------------------|-------------|------------------------|
| default   | task-manager-service |             | http://127.0.0.1:37895 |
|-----------|----------------------|-------------|------------------------|
```

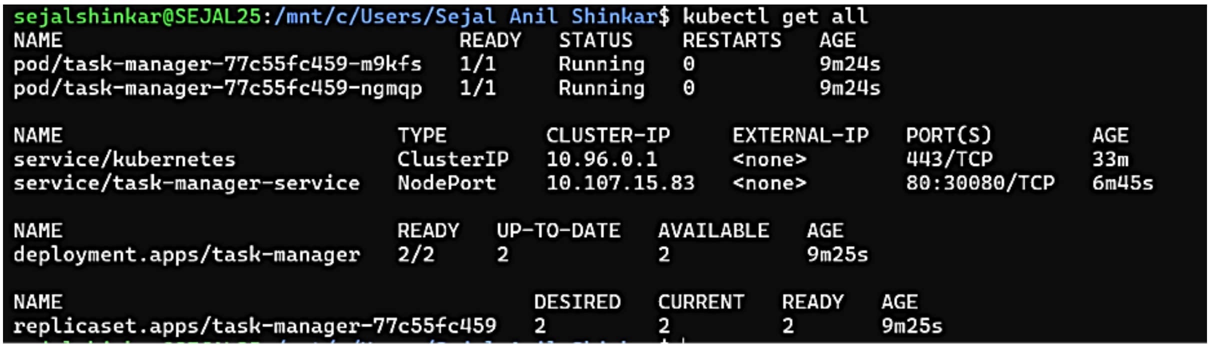


Student Task Management System

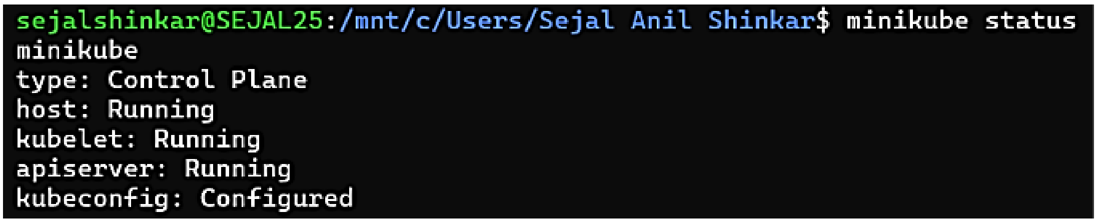
Keep track of your daily college work

My Tasks

No tasks added yet.



```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar$ kubectl get all
NAME                                  READY   STATUS    RESTARTS   AGE
pod/task-manager-77c55fc459-m9kfs     1/1     Running   0          9m24s
pod/task-manager-77c55fc459-ngmqp     1/1     Running   0          9m24s

NAME                           TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/kubernetes             ClusterIP   10.96.0.1       <none>        443/TCP        33m
service/task-manager-service   NodePort    10.107.15.83    <none>        80:30080/TCP   6m45s

NAME                           READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/task-manager   2/2     2            2           9m25s

NAME                                      DESIRED   CURRENT   READY   AGE
replicaset.apps/task-manager-77c55fc459   2         2         2       9m25s
```



```
sejalshinkar@SEJAL25:/mnt/c/Users/Sejal Anil Shinkar$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```